Executive Summary: NBA Playoff/Championship Model

This project develops a predictive modeling framework to estimate each NBA team's probability of winning a championship this season using team-level and aggregated player performance metrics. The objective is to generate probabilistic estimates that reflect each team’s relative likelihood of success.

The dataset will consist of multiple NBA seasons (most likely a sample of 5-10 seasons), with each observation representing a team-season. The response variable will be defined as a binary indicator of championship outcome; however, the primary focus of the model will be on estimating the conditional probability of winning given a set of predictors. This approach allows for more informative comparisons across teams and aligns with real-world forecasting methods.

The feature space will include a rich set of team-level predictors such as offensive and defensive efficiency, point differential, pace, turnover rate, rebounding rate, and shooting efficiency metrics. In addition, the model will incorporate derived and interaction features such as offense–defense interactions, pace-adjusted efficiencies, and measures of roster composition (e.g., top-player impact and team depth) to capture nonlinear relationships and team dynamics. This expanded feature set creates a high-dimensional setting with potentially dozens to hundreds of predictors.

To address this, the project will employ regularized statistical learning methods, specifically LASSO-based logistic models, to estimate the relationship between predictors and postseason success. Regularization will enable variable selection and coefficient shrinkage, improving model interpretability while mitigating overfitting in the presence of many correlated features.

The modeling approach is also informed by prior work in basketball analytics, particularly adjusted plus-minus frameworks that estimate player contributions using regression-based methods. Notably, Joseph Sill introduced Regularized Adjusted Plus-Minus (RAPM), which applies ridge regression to address multicollinearity and noise in player evaluation. Foundational concepts from basketball analytics literature, such as those developed by Dean Oliver, also guide the selection of efficiency-based team metrics.

The model will be evaluated using out-of-sample prediction techniques, such as train-test splits or cross-validation, with performance assessed through appropriate metrics (e.g., accuracy, log-loss, or classification error). In addition to predictive performance, the analysis will examine which team characteristics are most strongly associated with deep playoff runs, providing insight into the factors that drive championship success.

Overall, this project applies high-dimensional statistical modeling techniques to a sports analytics context, offering both predictive insights and an interpretable framework for understanding team performance in the NBA.

References

- Robert Tibshirani (1996). *Regression Shrinkage and Selection via the Lasso*.
- Joseph Sill (2010). *Improved NBA Adjusted Plus-Minus Using Regularization*.
- Dean Oliver (2004). *Basketball on Paper*.